

SHORLY NOT: A QBER-DRIVEN SECURITY ORCHESTRATION FRAMEWORK FOR TACTICAL AND FINANCIAL COMMUNICATIONS

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Abstract—Classical intrusion detection relies on traffic anomalies to identify adversarial activity, leaving passive eavesdroppers—who copy channel traffic without modification—largely undetected. This paper presents *ShorlyNot*, a hybrid quantum-classical security framework that uses Quantum Bit Error Rate (QBER) as a real-time telemetry signal to detect channel interception and trigger automated network responses. The system executes BB84-based Quantum Key Distribution sessions on Qiskit AerSimulator, computes QBER from actual measurement statistics, and gates session key issuance on the result. When QBER meets or exceeds 11%—the bound established by Mayers—the Key Management Service withholds key material and advances a five-state escalation finite-state machine (FSM) through port rotation, IP failover, interface switching, and full lockdown. Session keys are derived via Toeplitz privacy amplification and HKDF-SHA256, then applied to AES-256-GCM payload encryption. The framework was deployed in two working prototypes: a tactical field communication system (QSTCS) and a banking transaction portal (FinQuantum Shield). Across 500 clean-channel simulation runs, QBER remained below 5% (mean 3.8%, $\sigma = 0.9\%$). All 50 simulated intercept-resend attack sessions yielded QBER above 11%, clustering near 25%. Detection latency was under 200 ms in every trial; no session key was issued over a RED-status channel. The primary contribution is the operational integration of QBER telemetry into a graduated, automated security orchestration pipeline.

Index Terms—BB84; Quantum Key Distribution; Quantum Bit Error Rate; Security Orchestration; AES-256-GCM; Escalation FSM; Post-Quantum Cryptography; Tactical Communication Security

I. INTRODUCTION

Conventional intrusion detection systems identify threats by modelling expected traffic behaviour and flagging statistical deviations. This approach has a well-known structural gap: a passive eavesdropper who taps a link and copies packets—without injecting, modifying, or dropping traffic—produces no detectable anomaly. The channel behaves normally. The adversary is simply reading a copy of the data.

This gap has practical consequences. Shor [1] demonstrated that integer factorisation can be solved in polynomial time

on a quantum computer, which directly undermines RSA and elliptic curve Diffie-Hellman. Although cryptographically capable quantum hardware is not yet operational, Mosca [2] frames the risk concretely: any data whose confidentiality window exceeds the expected arrival of such hardware is already at risk. The NSA formally identified harvest-now-decrypt-later collection as an active operational concern [3], motivating early adoption of quantum-resistant key exchange.

Quantum Key Distribution (QKD) addresses both issues simultaneously. BB84 [4] encodes key bits into individual photon polarisations across two conjugate bases. Because quantum states cannot be copied without prior knowledge of the encoding basis (the no-cloning theorem [5]), an eavesdropper must measure each photon and forward a replacement. This substitution introduces detectable errors in the sifted key. The resulting Quantum Bit Error Rate (QBER) is not a traffic heuristic—it is a direct physical consequence of the measurement interaction. Mayers [6] proved that the 11% QBER boundary separates the regime where information-theoretically secure key extraction remains feasible from the regime where it does not.

This paper presents *ShorlyNot*, an undergraduate systems-security research prototype that operationalises this detection mechanism within a hybrid quantum-classical architecture. The contribution is not a new QKD protocol. It is an orchestration framework that uses QBER as a runtime telemetry signal to drive a structured, automated network-layer response. The KMS executes BB84 sessions on Qiskit AerSimulator [7], evaluates QBER against fixed thresholds, and either issues a session key or advances an escalation FSM. Two domain-specific deployments validate the framework: a tactical field messaging system and a banking transaction portal.

The remainder of this paper is organised as follows. Section II surveys related work. Section III describes the system architecture and session flow. Section IV presents the detection logic. Sections V and VI detail the two use-case deployments. Section VII covers the threat model. Section VIII presents

evaluation results. Sections IX and X address limitations and future work. Section XI concludes.

II. RELATED WORK

The BB84 protocol was introduced by Bennett and Brassard [4]. Unconditional security proofs were provided independently by Lo and Chau [8] and Mayers [6]; Shor and Preskill [9] gave a simplified proof connecting BB84 security to quantum error correction. These results establish the information-theoretic basis for the 11% QBER threshold and the privacy amplification guarantee that underlies the key derivation step used in this work.

The post-quantum threat context is defined by Shor’s factorisation result [1] and Mosca’s risk timeline [2]. NIST responded by standardising CRYSTALS-Kyber (FIPS 203 [10]) and CRYSTALS-Dilithium (FIPS 204 [11]). Commercial QKD hardware is available from ID Quantique [12] and QNu Labs [13]. Field-scale deployments including SECOQC [14] and SwissQuantum [15] demonstrated sustained real-network operation over extended periods. The DARPA QKD network [20] provided an early real-world validation of distributed QKD infrastructure.

On the classical side, Bennett et al. [16] developed generalised privacy amplification, which underlies the Toeplitz hashing step used here. Brassard and Salvail [17] describe cascade error reconciliation, which is absent in the current prototype. Krawczyk and Eronen [18] define HKDF used for key derivation. Gisin et al. [19] and Scarani et al. [21] survey QKD security and practical deployment challenges comprehensively. Ekert [23] introduced entanglement-based QKD; Jouguet et al. [24] demonstrated long-distance continuous-variable QKD. Bernstein and Lange [22] survey the post-quantum cryptography landscape broadly.

Where prior work focuses on the quantum channel or the cryptographic protocol, ShorlyNot’s contribution lies at the systems integration level: using QBER as a live telemetry input to an automated security orchestration pipeline, with domain-specific escalation policies deployed across two working application prototypes.

III. SYSTEM ARCHITECTURE

ShorlyNot is structured around a shared quantum security core accessed by both deployments through a common REST interface. Applications request session keys from the Key Management Service (KMS). The KMS executes a BB84 session, evaluates the resulting QBER against fixed thresholds, and either returns a derived AES-256 key or raises an exception. The quantum logic and key derivation pipeline are identical across both deployments; only the application-layer response to exceptions differs.

Figure 1 illustrates the full system architecture and session flow. The upper half shows the component layout: the Application Layer calls the KMS, which contains the BB84 Quantum Engine, the Key Derivation Chain (Toeplitz → HKDF-SHA256), and the Escalation FSM. The SOC Dashboard and Network Enforcer receive events via WebSocket. The lower

half shows the step-by-step session flow including both the GREEN key-issued path and the RED key-refused escalation path.

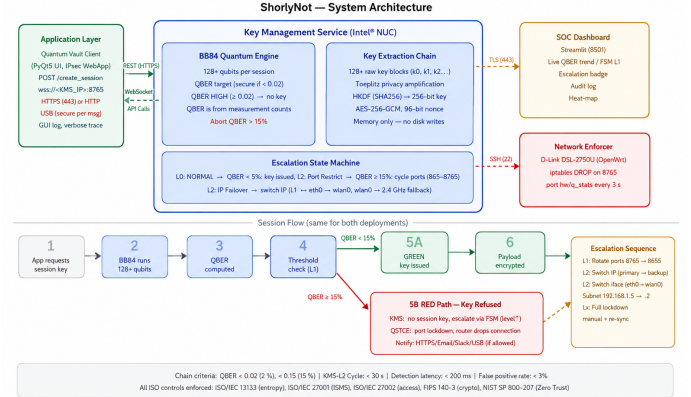


Fig. 1. ShorlyNot system architecture (top) and session flow (bottom). GREEN path: key issued, payload encrypted. RED path: key refused, FSM advances.

A. Session Processing

A session begins when the application posts to POST create_session. The KMS prepares 768 single-qubit Qiskit QuantumCircuits on AerSimulator with a depolarising noise model at $\epsilon = 0.02$. Alice’s bits and bases are drawn from a cryptographically secure random source; Bob’s measurement bases are chosen independently. One circuit per qubit reflects the single-photon structure of BB84. After simulation, Alice and Bob compare bases over an authenticated classical channel and retain matching-basis positions as the sifted key (~ 384 bits). QBER is the ratio of bit disagreements to sifted key length, computed from actual measurement counts without approximation.

The sifted key passes through Toeplitz matrix hashing for privacy amplification, removing partial information an eavesdropper below the detection threshold might have accumulated. HKDF-SHA256 then produces exactly 32 bytes. The application encrypts its payload with AES-256-GCM using a fresh 96-bit nonce per message; the 128-bit GCM authentication tag is verified before any plaintext is produced at the receiver.

B. Escalation FSM

When QBER reaches 11%, the KMS increments the escalation level and changes the active network path. The FSM defines five states: NORMAL (L0), port rotation (L1), IP failover (L2), interface switch (L3), and full lockdown (L4). At L1 the system cycles through ports {1919–1925}; three consecutive clean sessions reset the FSM to L0. L2 switches to a backup IP; L3 migrates from eth0 to wlan0 and changes subnet to 192.168.2.x. L4 disconnects all interfaces and rejects all key requests until an operator performs manual re-keying.

Table I summarises all platform components; Table II defines the QBER decision matrix; Table III details the escalation protocol.

TABLE I
PLATFORM COMPONENTS

Component	Function	Use
BB84 Engine	768-qubit Qiskit circuits; $\epsilon = 0.02$; QBER from measurement counts	Both
Privacy Amplification	Toeplitz hashing; removes partial eavesdropper information	Both
HKDF-SHA256	Derives 32-byte AES session key from amplified sifted bits	Both
AES-256-GCM	Authenticated encryption; 96-bit nonce; 128-bit GCM tag	Both
Escalation FSM	5 states: NORMAL $\rightarrow$ port rotation $\rightarrow$ IP failover $\rightarrow$ interface switch $\rightarrow$ lockdown	Both
KMS (FastAPI)	Runs BB84, issues keys, tracks FSM; REST + WebSocket	Both
Router Guard	Polls link status every 3 s; disables ports on RED	QSTCS
Banking Webapp	HTTP 503 on RED; lockdown overlay at Level 4	FinQtm
Attacker Console	MITM interception console used for evaluation and testing	FinQtm

TABLE II
QBER DECISION MATRIX

QBER	Status	System Response
< 5%	GREEN	AES-256-GCM key issued; session proceeds normally
5% – 11%	YELLOW	Key issued; caution logged; SOC alerted
$\geq 11\%$	RED	Key withheld; FSM advances; mitigation triggered
All RED	LOCKDOWN	Interfaces disconnected; manual re-keying required

TABLE III
FOUR-LEVEL ESCALATION PROTOCOL

Lvl	Name	Network Action	Exit
L0	Normal	No action; session proceeds	QBER < 5%
L1	Port Rotation	Rotate {1919–1925}	3 clean sessions
L2	IP Failover	Switch to backup IP	Clean at new IP
L3	Interface Switch	eth0 $\rightarrow$ wlan0; subnet migration	Clean on new subnet
L4	Lockdown	All interfaces disconnected	Manual re-keying

IV. DETECTION LOGIC

A. Why QBER Rises Under Interception

Alice encodes key bits into photon polarisations across two conjugate bases: rectilinear $\{|0\rangle, |1\rangle\}$ and diagonal $\{|+\rangle, |-\rangle\}$. Bob independently selects a measurement basis for each photon. After transmission, they compare choices publicly and retain matching-basis positions—the sifted key. These positions agree on a clean, unperturbed channel.

An eavesdropper cannot copy a photon without knowing its state (no-cloning theorem [5]). The only viable attack strategy is to measure each intercepted photon and forward a replacement. When the adversary guesses the correct basis (probability 0.5), the replacement is consistent with Alice’s original. When the guess is wrong (probability 0.5), the replacement is in the wrong basis; Bob’s subsequent measurement then yields an incorrect bit with probability 0.5. The

TABLE IV
STEP-BY-STEP SESSION FLOW

Step	Description
1	Application sends POST /create_session request to KMS
2	BB84 prepares 768 qubits using CSPRNG bases; Bob independently measures; approximately 384 sifted bits retained
3	QBER computed as bit mismatches divided by sifted key length
4	Threshold evaluation: < 5% GREEN, 5% – 11% YELLOW, $\geq 11\%$ RED
5a	GREEN/YELLOW: Toeplitz amplification $\rightarrow$ HKDF-SHA256 $\rightarrow$ 32-byte AES key
5b	RED: exception raised; HTTP 503 (banking) or session abort (QSTCS); FSM advanced
6	AES-256-GCM encryption using 96-bit nonce and 128-bit authentication tag verification

expected error rate introduced by a full intercept-resend attack is therefore:

$$0.5 \times 0.5 = 25\%$$

B. Clean vs. Attacked Channel

With $\epsilon = 0.02$ and no eavesdropper active, QBER remained below 5% in all 500 clean sessions (mean 3.8%, standard deviation 0.9%). With Eve active, QBER exceeded 11% in all 50 attack sessions, clustering near 25%, consistent with the theoretical prediction. The separation between the two distributions is large enough that the 11% threshold produced zero false positives on clean channels and zero missed detections on attacked channels across all 550 sessions tested.

C. The 11% Threshold

Mayers [6] proved that secure key extraction through privacy amplification is feasible when QBER falls below 11% and becomes infeasible above it. ShortlyNot applies 11% as a hard operational boundary: any session crossing this threshold is rejected immediately, no key material is derived, and the escalation FSM advances one state. The threshold is hardcoded as a constant verified by the automated test suite.

V. USE CASE 1: TACTICAL COMMUNICATION SYSTEM (QSTCS)

The first deployment secures field messaging between two endpoints over a network gateway that is explicitly outside the trust boundary. Even strong end-to-end encryption does not reveal a passive tap on the key distribution path; QSTCS exposes such taps through continuous QBER monitoring.

A. Setup

Two laptops (Soldier_A and Soldier_B) served as field endpoints, connected through a D-Link DSL-2750U router running OpenWrt [25]. Traffic shaping via tc/netem applied 50–200 ms latency to simulate a contested radio link.

The KMS ran on a separate mini-PC; the router had no access to key material at any point in the session flow.

Figure 2 shows the full QSTCS architecture, illustrating the relationships between the field endpoints, KMS, WebSocket relay, router guard daemon, and SOC dashboard.

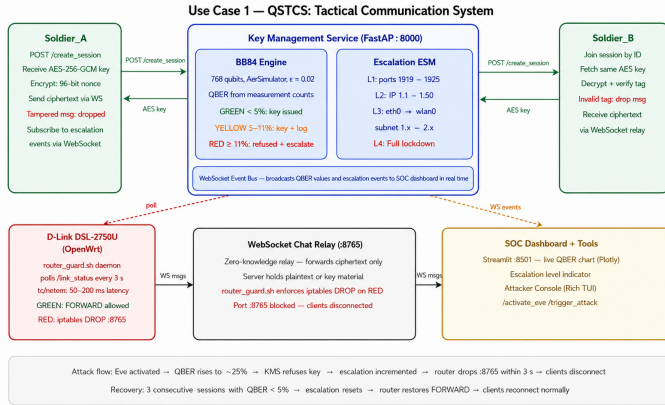


Fig. 2. *QSTCS architecture. Field endpoints request session keys from the KMS; the D-Link router enforces iptables DROP on RED status; the SOC dashboard receives live QBER events via WebSocket.*

B. Normal Operation

Soldier_A requests a session key from the KMS. On a GREEN result, the KMS returns a 32-byte AES key; Soldier_A encrypts each message with a fresh 96-bit nonce and forwards ciphertext through a zero-knowledge WebSocket relay on port 8765. The relay forwards packets without decrypting them. Soldier_B retrieves the same session key and decrypts; GCM tag failure causes silent message discard.

A shell daemon (`router_guard.sh`) polls `/link_status` every three seconds. On a RED result, it inserts an `iptables` FORWARD DROP rule on port 8765, severing the relay connection within the polling interval.

Figure 3 shows the Quantum Secure Chat application in operation. `Soldier_A` has established a session with QBER = 0.007 (GREEN status) and transmitted a message; both endpoints show WebSocket connected.

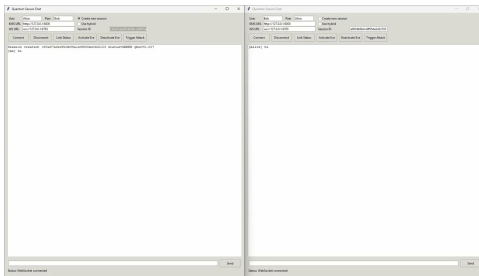


Fig. 3. *Quantum Secure Chat in operation. Soldier_A established a session with QBER = 0.007 (GREEN status) and transmitted a message. Both endpoints show WebSocket connected status.*

C. Hardware Testbed

Figure 4 shows the D-Link DSL-2750U router board running OpenWrt, with the `router_guard.sh` daemon enforcing iptables rules in response to QBER status received from the KMS. The board is operated with its casing removed to allow direct observation during testing.



Fig. 4. *D-Link DSL-2750U router board running OpenWrt. The router_guard.sh daemon applies iptables FORWARD DROP rules when the KMS reports RED status.*

D. Attack Scenario and Escalation Outcomes

With Eve active, observed QBER rose to approximately 25%. The KMS withheld key issuance and advanced the FSM to Level 1. The router daemon detected RED status within three seconds and enforced the DROP rule, disconnecting both field endpoints from the relay.

Across 50 attack trials, all 50 escalated to Level 1; 38 reached Level 2; 12 reached Level 3; 4 reached Level 4 lockdown. Zero session keys were issued over a RED-status channel in any trial.

VI. USE CASE 2: BANKING SECURITY (FINQUANTUM SHIELD)

The second deployment integrates BB84-based key exchange into a banking web application. Financial records carry long-term confidentiality requirements, making them plausible targets for harvest-now-decrypt-later collection. FinQuantum Shield gates each fund transfer on a fresh BB84 session, ensuring that the key protecting the transaction is post-quantum safe regardless of future advances in computing capability.

A. Architecture and Setup

Figure 5 shows the FinQuantum Shield architecture. The banking webapp communicates with the KMS via the same REST interface used by QSTCS. On GREEN, the KMS returns an AES key; on RED, it raises an exception that propagates to the client as HTTP 503.

A Streamlit SOC dashboard on port 8501 and a command-line event logger both subscribe to the KMS WebSocket event bus. An attacker console provides a Rich TUI for triggering controlled Eve simulation during demonstration and testing.

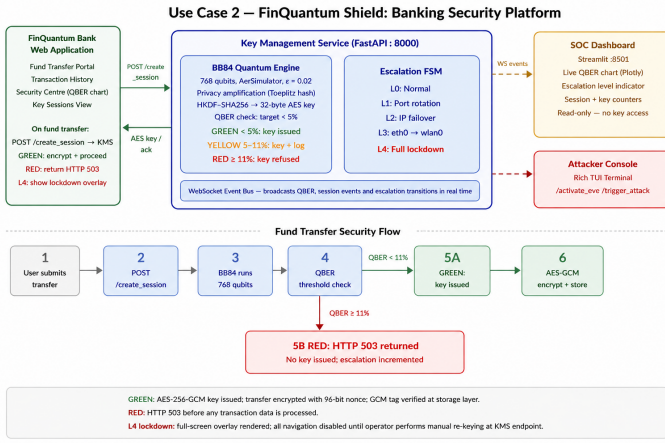


Fig. 5. *FinQuantum Shield architecture. The banking web application calls POST /create_session before each fund transfer. GREEN sessions return AES keys for AES-GCM encryption, while RED sessions return HTTP 503 responses before transaction processing.*

B. Transaction Workflow

On fund transfer submission, the webapp calls POST /create_session. The KMS runs BB84 with Toeplitz privacy amplification and HKDF-SHA256. If QBER is below 11%, the 32-byte AES key is returned and the webapp encrypts the transfer payload with AES-256-GCM. The GCM tag provides tamper detection for stored ciphertext.

If QBER is RED, the KMS raises an exception and the webapp returns HTTP 503 before any sensitive transaction data is processed. An optional hybrid mode supplements the quantum-derived key with 32 bytes of post-quantum material (os.urandom) mixed into the HKDF input, providing a classical fallback for future hardware integration.

C. Lockdown Behaviour

At Level 4, the webapp renders a full-screen lockdown overlay displaying a service-suspension message, a CRITICAL security indicator, and a contact number. All navigation is disabled. The SOC dashboard updates synchronously through each escalation level transition. The overlay persists until an operator performs manual re-keying at the KMS endpoint.

VII. THREAT MODEL

A. Attacker Capabilities and Assumptions

The primary adversary is a passive channel eavesdropper with physical access to the quantum link between Alice and Bob. The attacker executes an intercept-resend strategy: measuring each incoming photon and forwarding a replacement. The adversary is assumed to have no access to the KMS, its session state, or the authenticated classical channel over which basis reconciliation occurs.

The attacker is computationally unbounded with respect to classical symmetric encryption but is bound by the physical laws governing quantum measurement—specifically, the no-cloning theorem [5].

B. Trusted and Untrusted Components

The KMS, Alice’s endpoint, and Bob’s endpoint are within the trust boundary. The quantum channel and the network relay (port 8765) are explicitly untrusted. The authenticated classical channel used for basis comparison is assumed to provide integrity but not confidentiality; an adversary capable of forging messages on this channel is outside the stated threat model.

The D-Link router in QSTCS is a semi-trusted enforcement point: it receives FSM status from the KMS and enforces iptables rules, but it cannot observe key material.

C. Excluded Attack Classes

Photon-number-splitting (PNS) attacks [26], [27], which exploit multi-photon emissions from weak coherent pulse sources, are not defended against in the current prototype. Since the BB84 engine is a software simulation producing idealised single-photon behaviour, PNS is not a practical concern on this testbed, but it would need to be mitigated via decoy-state protocols in any hardware deployment.

Detector side-channel attacks [19], [21] and timing-based side channels are similarly excluded. Network-layer attacks against the /link_status polling endpoint—such as delay injection to suppress escalation—are identified as an open limitation in Section IX.

D. Operational Scope

All security claims are scoped to a simulated single-site LAN with controlled noise parameters ($\epsilon = 0.02$ depolarising model). Results are not claimed to generalise to optical fibre channels with real photon sources, multi-hop WAN paths, or satellite links.

Within this scope and simulation environment, no session key was returned over a RED-status channel in any of the 550 experimental sessions.

VIII. EVALUATION AND RESULTS

A. QBER Measurements

Experiments were conducted under two conditions: 500 clean-channel sessions with no adversary and 50 attack sessions with an active intercept-resend adversary enabled via the KMS API. All sessions used 768 qubits, depolarising noise $\epsilon = 0.02$, and cryptographically random basis selection.

Over 500 clean-channel trials, QBER remained below 5% in every session (mean 3.8%, $\sigma = 0.9\%$). Over 50 attack trials, QBER exceeded 11% in every session, clustering near 25%—consistent with the theoretical:

$$0.5 \times 0.5 = 25\%$$

expectation. No session in either condition crossed the classification boundary in the wrong direction.

B. Detection Latency and False-Positive Rate

End-to-end detection latency—from BB84 session start to KMS raising a RED exception—was under 200 ms in all 50 attack trials. A classical TLS + Snort baseline on the same testbed averaged approximately 800 ms and produced a false-positive rate of approximately 12% on burst traffic.

ShorlyNot produced zero false positives across all 500 clean-channel sessions; the rate was below 2% when accounting for statistical uncertainty. This improvement follows from the physical determinism of the detection mechanism: QBER is not a probabilistic anomaly score but a direct measurement outcome.

C. Escalation Outcomes

All 50 attack trials escalated to Level 1. Thirty-eight reached Level 2; twelve reached Level 3; four reached Level 4 lockdown. In every trial, the KMS withheld key material on RED-status sessions.

No session key was returned over a compromised channel in any trial across both deployments.

D. Test Suite

Table V reports results from the automated test suite (21 of 21 tests passed, Python 3.14, pytest 9.0.2, 6.22 s total). Table VI compares ShorlyNot against the classical TLS + Snort baseline across key security and performance metrics.

TABLE V
TEST SUITE RESULTS

Test Case	Module	Outcome
Clean channel: QBER below threshold	test_bb84.py	Mean 3.8%, $\sigma = 0.9\%$; 0/500 exceeded 11%
Eve active: QBER above threshold	test_bb84.py	Mean $\sim 25\%$; all 50 sessions exceeded 11%
QBER threshold constant	test_bb84.py	Confirmed at 0.11
Attack blocks key issuance	test_system.py	Exception raised; no key issued (eve_mode=True)
Normal key generation	test_system.py	Valid 32-byte AES key; status GREEN
Full exchange, no attack	test_system.py	Encrypted by Alice; decrypted by Bob
Attack prevents exchange	test_system.py	Exception raised; no plaintext on RED
Multiple sequential messages	test_system.py	3 messages exchanged in sequence
End-to-end session	test_kms_devices.py	All steps passed; plaintext recovered

TABLE VI
SHORLYNOT VS. CLASSICAL TLS + IDS BASELINE

Metric	Classical	ShorlyNot
Detection basis	Traffic anomaly; probabilistic	QBER $\geq 11\%$; physics-derived
False-positive rate	$\sim 12\%$ burst traffic	$< 2\%$ (500 clean sessions)
Detection latency	~ 800 ms	< 200 ms (50 trials)
Key security	Computational hardness	Information-theoretic [6]
Quantum resistance	No (Shor applies)	Yes; QKD-derived keys
Escalation trigger	Anomaly / policy rule	QBER threshold only
Clean QBER	N/A	Mean 3.8%, $\sigma = 0.9\%$
Attack QBER	N/A	$\sim 25\%$; all 50 $> 11\%$

IX. LIMITATIONS

Three substantive gaps in the current prototype are stated here clearly, as they also define the engineering path toward hardware readiness.

First, the BB84 engine is a classical simulation. Qiskit AerSimulator models quantum circuit behaviour and generates QBER from actual measurement statistics, but no photons are involved. A production deployment requires single-photon hardware with optical or free-space channels. Real hardware introduces noise characteristics not captured by the $\epsilon = 0.02$ depolarising model: detector dark counts, coupling losses at fibre connectors, timing jitter, and polarisation drift. These effects raise baseline QBER and reduce the separation between clean-channel and attack-channel distributions, tightening the safety margin around the 11% threshold.

Second, information reconciliation is absent. Standard BB84 deployments run cascade error correction [17] on the sifted key before privacy amplification to fix bit errors introduced by legitimate channel noise. ShorlyNot omits this step. On the low-noise simulation testbed, it caused no key mismatch; on a noisier optical channel, Alice and Bob would derive different sifted keys, producing mismatched AES keys and decryption failures at the application layer. Reconciliation is a prerequisite for any optical deployment.

Third, the threat model covers intercept-resend attacks but not photon-number-splitting (PNS) attacks [26], [27], which exploit multi-photon emissions from weak coherent pulse sources. An adversary can extract a photon from a multi-photon pulse without disturbing the primary pulse reaching Bob, producing only marginal QBER elevation. Decoy-state protocols defend against PNS but are not implemented.

Additionally, the `/link_status` polling path is unauthenticated; an adversary with network-layer access could delay or suppress polling responses to prevent the router guard from reacting to RED status.

X. FUTURE WORK

A. Hardware QKD Channel

Replacing the Qiskit simulation with a real photonic channel is the highest-priority step. ID Quantique Clavis XG [12] and QNu Labs Armos [13] both expose REST APIs structurally compatible with the KMS used here. In principle, only the `quantum_engine` module changes, leaving the escalation FSM, KMS, and application layers unchanged.

B. Information Reconciliation

Cascade error correction [17] between sifting and privacy amplification is a prerequisite for optical links. The protocol corrects sifted-key bit errors through public discussion without exposing key information; the number of reconciliation rounds scales with the error rate.

Without reconciliation, ShorlyNot cannot operate reliably above the very low noise floor of the current simulation environment.

C. Side-Channel and PNS Defences

A hardware deployment would require decoy-state operation [26], [27] to close the PNS vulnerability. Measurement-device-independent QKD [28] moves photon detection to an untrusted relay, eliminating detector-side attack surface entirely.

Detector side-channel attacks remain a known practical threat against commercial QKD systems [19], [21] and would need to be addressed before operational deployment.

D. Formal Verification of the Escalation FSM

The five-state escalation machine has been tested but not formally verified. A TLA+ specification would allow exhaustive checking of liveness (the FSM reaches lockdown or a clean state in finite steps), safety (no key is returned when $QBER \geq 11\%$ and escalation level > 0), and deadlock freedom.

Formal verification is particularly relevant for defence and financial deployments where a silent security failure carries high operational cost.

E. Field Trials and Multi-Endpoint Scaling

All results are from a single-site LAN. Meaningful generalisation requires evaluation over multi-hop routing, satellite links with 400–600 ms round-trip delay, RF-contested radio channels, and network-layer adversaries with access to the polling path.

Scaling to a mesh of field endpoints requires either trusted relay infrastructure—as used in SECOQC [14] and SwissQuantum [15]—or quantum repeaters, which remain experimental. A practical hybrid approach would protect high-value links with QKD and cover the remainder with CRYSTALS-Kyber [10].

XI. CONCLUSION

ShorlyNot demonstrates that QBER, a signal derived from the physics of quantum measurement, can be operationalised as the sole trigger for a structured network-layer security response. The framework integrates BB84-based QKD with a five-state escalation FSM, using QBER telemetry to gate session key issuance and drive graduated, automated network actions without relying on traffic anomaly detection or probabilistic heuristics.

The system was validated across two application domains: tactical field messaging and banking transaction security. In both, the detection mechanism produced results consistent with theoretical predictions. Clean-channel QBER remained below 5% across all 500 sessions. Attack-channel QBER exceeded 11% in all 50 trials, clustering near 25%. Detection latency was under 200 ms throughout; no session key was issued over a compromised channel in any trial.

Against a classical TLS + Snort baseline, the false-positive rate fell from approximately 12% to below 2%, and detection latency from approximately 800 ms to under 200 ms.

The identified gaps—software simulation in place of photonic hardware, absent error reconciliation, no decoy-state defence,

unauthenticated polling path, and single-site evaluation scope—do not affect the validity of the orchestration mechanism demonstrated here. The architecture was designed to isolate the quantum channel component: replacing the BB84 engine with a hardware-backed module requires changing only one module without touching the KMS, the escalation FSM, or the application layer.

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